

Research article

Recognizing the state of aerobic granular sludge over its life-cycle in a continuous-flow membrane bioreactor with an artificial intelligence approach

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ABSTRACT

The continuous-flow aerobic granular sludge-membrane bioreactor (AGS-MBR) system represents an efficient and sustainable technology for wastewater treatment. AGS, a spherical or ellipsoidal granular sludge formed through microbial self-aggregation under aerobic conditions, progresses through four distinct life-cycle stages in the AGS-MBR system: initial, growth, mature, and cleaved. Accurate identification and classification of these stages are crucial for optimizing AGS-MBR operations and maintaining system stability; however, traditional monitoring methods are labor-intensive and error-prone. This study utilized Artificial Intelligence (AI) to develop a machine learning model based on the You Only Look Once (YOLOv8) algorithm for automated AGS monitoring and classification. Trained on 862 annotated images, the model achieved average precision of 0.985 at an Intersection over Union (IoU) threshold of 0.5 (mAP50), and the mAP50-95 of 0.837, demonstrating high accuracy in AGS classification. The t-distributed Stochastic Neighbor Embedding (t-SNE) revealed distinct clusters of AGS features across life-cycle stages, while SHapley Additive exPlanations (SHAP) demonstrated that the model focused on global features of small-grained images and edge features of large-grained images, both confirming the robustness of classification. The model's statistical functionality, supported by global variables, enabled real-time AGS monitoring in MBR system. This study provides a powerful tool for detecting and classifying the AGS life-cycle, offering guidance for the operation and maintenance of AGS-MBR system and demonstrating the potential applications of AI in wastewater treatment.

1. Introduction

Aerobic granular sludge (AGS), characterized by its unique structure and high biomass density, offers significant advantages in the simultaneous removal of nutrients and organic pollutants within a single bioreactor, potentially surpassing the conventional activated sludge (CAS) process in terms of efficiency and cost-effectiveness (Bengtsson et al., 2019; Hu et al., 2024). The long-term stability of AGS is pivotal to sustaining these benefits, while the sequencing batch reactor (SBR) in batch mode has been the most successful method for AGS cultivation (Han et al., 2024), there is a burgeoning interest in continuous mode cultivation, especially when AGS is integrated with membrane bioreactors (MBRs). MBRs, which combine CAS with membrane filtration, are known for producing high-quality effluent and reducing wasted

activated sludge (WAS) (Rahman et al., 2023). However, conventional MBRs often face issues with membrane fouling due to the deposition of flocculated activated sludge particles on the membrane surface (Morales et al., 2024). In contrast, AGS, with its aggregate form, has demonstrated an effective mitigation of membrane fouling, which is crucial for enhancing the practicality of MBR applications (Campo et al., 2021). Therefore, the AGS-MBR system stands out as an efficient, economical, and sustainable wastewater treatment technology that not only minimizes membrane fouling but also ensures superior effluent quality and reduced WAS production.

In AGS-MBR system, the stability of AGS is paramount, with morphological integrity and structure being key to operational efficiency. The AGS life-cycle, encompassing initial, growth, mature, and cleaved stages, significantly influences bioreactor performance (Yang

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et al., 2023). Monitoring AGS attributes, such as morphology, structure, and size distribution, is essential for effective process control (de Sousa Rollemberg et al., 2018). Then, operational parameters like influent composition and temperature can induce variability in AGS, potentially leading to a transition from granular to flocculated sludge and system instability (Iorhemen and Liu, 2021). The self-cultivation of AGS in continuous flow MBR system has gained interest due to operational convenience and enhanced performance. However, the heterogeneity of AGS, with varying physicochemical, microbiological, and morphological properties, remains a challenge (Yang et al., 2023). In the initial stage, the sludge particles are small; in the growth stage, the sludge particles have rough edges; in the mature stage, the sludge particles have a stable structure and smooth surface; in the cleaved stage, the sludge particles age, with a decrease in large particles and an increase in small ones. The characteristics of each stage are interrelated, reflecting the dynamic changes of AGS from formation to aging. Understanding the differences and features of these life-cycles is of great significance for optimizing the operation and management of AGS-MBR system, which can help us better control and adjust operational conditions to maintain the stability of AGS and improve the efficiency of wastewater treatment. In addition, this information also provides an important reference for the design of new bioreactors or the improvement of existing technologies.

Accurate state identification of AGS is crucial, yet methods like frozen section analysis, dissolved oxygen (DO) micro-electrodes, and CLSM have limitations. Invasive techniques can disrupt the internal micro-environment, while CLSM requires laborious sample preparation. These approaches may also be sensitive to AGS properties, leading to inaccuracies and a lack of reproducibility (Kirkland et al., 2020; Li et al., 2014; Ranzinger et al., 2020). This underscores the uniqueness of AGS in MBR operations and the necessity for advanced diagnostic tools for its state assessment, which is vital for optimizing the performance of AGS-MBR system. The development of intelligent, non-invasive methods to analyze AGS states in real-time is therefore imperative, ensuring the long-term stability and efficiency of these system.

Artificial intelligence (AI) has demonstrated its prowess in wastewater treatment modeling by offering enhanced learning capabilities, automation, and adaptability beyond traditional methods (Alvi et al., 2023; Narayanan et al., 2024). AI-driven models can reveal complex data relationships and adapt to new data, enhancing understanding of wastewater treatment processes and aiding in prediction and control (Ignacz et al., 2025; Pérez-Beltrán et al., 2024). Within the context of AGS system, artificial neural networks (ANNs) have been effectively utilized to predict critical process outcomes (Zhong et al., 2021), such as the removal rates of chemical oxygen demand, ammonia nitrogen, total nitrogen, and total phosphorus (Zaghloul and Achari, 2022). They have also been employed to predict sludge properties and water quality based on historical reactor data, enhancing the predictive accuracy and control of AGS processes (Mahmod and Wahab, 2017; Zaghloul et al., 2021). Despite these advancements, there is a significant gap in the application of AI for the image recognition and classification of aerobic granular sludge (AGS). You Only Look Once (YOLO) is a highly influential AI technology that, based on a deep learning framework, can efficiently identify the positions and categories of objects in images, demonstrating strong real-time performance and accuracy (Redmon et al., 2016). The YOLO algorithm has not only been widely applied in multiple fields such as agricultural disease monitoring (Gao et al., 2024), medical image diagnosis (Xu et al., 2025), infrastructure inspection (Santos and Carvalho, 2025), and autonomous driving (Ul Islam et al., 2025), but also provides strong support for machine learning in target recognition tasks in complex scenarios due to its efficiency and adaptability. It is an indispensable excellent tool in the field of image recognition. Perhaps the YOLO algorithm could be a new approach to achieving the image recognition and classification of AGS.

Based on the aforementioned background, this study aimed to develop a machine learning approach for accurately identifying the

states of AGS in a continuous-flow MBR over its entire life-cycle using the YOLOv8 object detection model. The model was trained following dataset screening, classification, and labeling, and then underwent hyperparameter tuning after pre-processing. Various evaluation metrics were calculated on the test set to quantify the model's predictive effectiveness, guiding the optimization of model parameters based on key indices. Additionally, dimensionality reduction techniques were applied to the semantic features of all AGS images in the test set, projecting them into two-dimensional visualizations. Feature clustering analysis was conducted across different categories, and interpretability analysis was performed to identify key features within images, elucidating differences and connections among them. Lastly, to facilitate real-world applications and enhance detection efficiency, a batch counting function was developed for each AGS stage, aiming to provide intelligent methods for obtaining timely statistical data to aid in the management and optimization of AGS-MBR processes.

2. Materials and methods

2.1. Description of the used AGS-MBR system

The study employed a previously reported MBR system with internal circulation as the physical model (Dai et al., 2020), in which, a multi-habitat environment was created through accumulated biomass, facilitated by the hydrodynamic conditions of internal circulation. The activated sludge used for inoculation was obtained from the secondary sedimentation tank of a local municipal wastewater treatment plant located in Haizhu District, Guangzhou, China (Biological Island Sewage Treatment Plant). The initial concentration of mixed liquor suspended solids (MLSS) was maintained at approximately 3000 mg/L. Throughout the experiment, the hydraulic retention time (HRT) ranged between 8.0 and 10.0 h, while the aeration rate varied from 4.8 to 7.2 m³/h to ensure sufficient DO supply. The AGS-MBR system used simulated wastewater as influent and consists of: CH₃COONa (769.20 mg/L), NH₄Cl (152.67 mg/L), KH₂PO₄ (26.36 mg/L), CaCl₂ (35.00 mg/L), FeSO₄ (0.85–0.95 mg/L), MgSO₄ (20.25 mg/L), NaHCO₃ (267.00 mg/L), corresponding COD, NH₄-N and TP concentrations of 600 mg/L, 40 mg/L and 6 mg/L. All other aspects related to system configuration and operational conditions remained consistent with the previously reported work (Dai et al., 2020).

2.2. Data collection and feature classification

In the 100-day operation process, the system transitioned from startup to stable operation. Sludge samples were collected daily at fixed locations and fixed times throughout this period. These samples were examined using bright field microscopy (Leica DM500, Wetzlar, Germany) at 40 × magnification. Timely images were taken with an attached digital camera, and some images were adjusted for depth of field to ensure clarity. To enhance model generalization ability and robustness, a total of 862 images were combined from this study and a previous work (Dai et al., 2020). For data acquisition, conforming data corresponding to each stage's characteristics in the experiment underwent initial pre-processing and classification into four distinct stages based on the morphological features and size variation of AGS, namely: initial stage (IS), growth stage (GS), maturity stage (MS), and cleavage stage (CS) (Yang et al., 2023). Specific classification rules and demonstrations were presented in Table 1 and Fig. 1.

In this study, the target detection annotation tool (Labelimg v1.8.6) was used to annotate the AGS image dataset across its life-cycle. The annotation included the location and centroid information of the targets, and the results were stored in TXT files. The dataset was divided into a training set (690 images) and a testing set (172 images) at a ratio of 80% and 20%. A random seed was set to ensure the reproducibility of the results.

The annotation process was as follows: First, a portion of the

Table 1
Classification of AGS whole life-cycle images.

Labels	Equivalent Size (μm)	Shape	Roughness	Color
IS	200–800	Irregular shape	Rough	Yellow, brown
GS	800–1600	Rough Circles	Rough	Yellow, brown, dark brown
MS	800–3000	Circles and ovals	Smooth	Dark brown, black
CS	1600–3000	Cracked shape	Rough	Dark brown, black

microscopic images from the sample set was selected for the training and testing sets. The images were imported into Labeling, and the target regions were selected using the mouse. The corresponding “YOLO” label categories were then chosen. When a single image contained multiple sludge particles, each particle was annotated according to its respective life-cycle stage. After the annotation was completed, the results were exported as TXT files. The images and their corresponding TXT files for the training and testing sets were then imported into their respective folders to complete the data preparation. Throughout the annotation process, manual annotation was strictly performed in accordance with the rules outlined in Table 1 to ensure annotation accuracy.

2.3. Machine learning model

In this study, YOLOv8 was utilized to extract supervised information from each image, including the precise location and category parameters of the detected objects. After all datasets were classified based on these criteria, an object detection dataset annotation tool (Labelimg

v1.8.6) was employed to set the center position, height, width, and class of each identified object to annotate the object position in each image. These results were then normalized, with the entire process illustrated in Fig. 2.

The Darknet-53 network is based on the use of residual and skip connections to effectively capture features at various scales and levels (Redmon and Farhadi, 2018), whose core idea can be described as the combination of a fundamental feature extraction framework and a residual module (Kaiming and Jian, 2014; Kaiming et al., 2015). In this study, it served as the foundation for extracting features from the input images (Redmon and Farhadi, 2017), where the network backbone employed a comprehensive gradient flow structure, with different channel numbers adjusted for diverse scale models. The neck section incorporated a bidirectional path network to introduce a bottom-up path that facilitated the transmission of low-level information to the top of the upper layers, while, in the head section, a contemporary decoupled head structure was adopted to distinguish between classification and detection heads. This approach enables direct prediction of the object centers instead of calculating offsets from predefined anchor frames. Additionally, for enhancing model precision and efficiency, this work integrated the Simple Attention Mechanism (SimAM) into the utilized network, enabling inference of three-dimensional attention weights within network layers without introducing additional parameters (Yang et al., 2021).

The image sizes were normalized in this study, followed by initialization of the model weights for training from scratch. Comprehensively considering both the results of model training and computational efficiency, the number of training iterations was set to 100 with a batch size of 16 images. The initial learning rate was set to 0.01 and gradually adjusted to achieve the minimum prediction error. To prevent overfitting, an early stop strategy was adopted (Hanwen et al., 2019), which

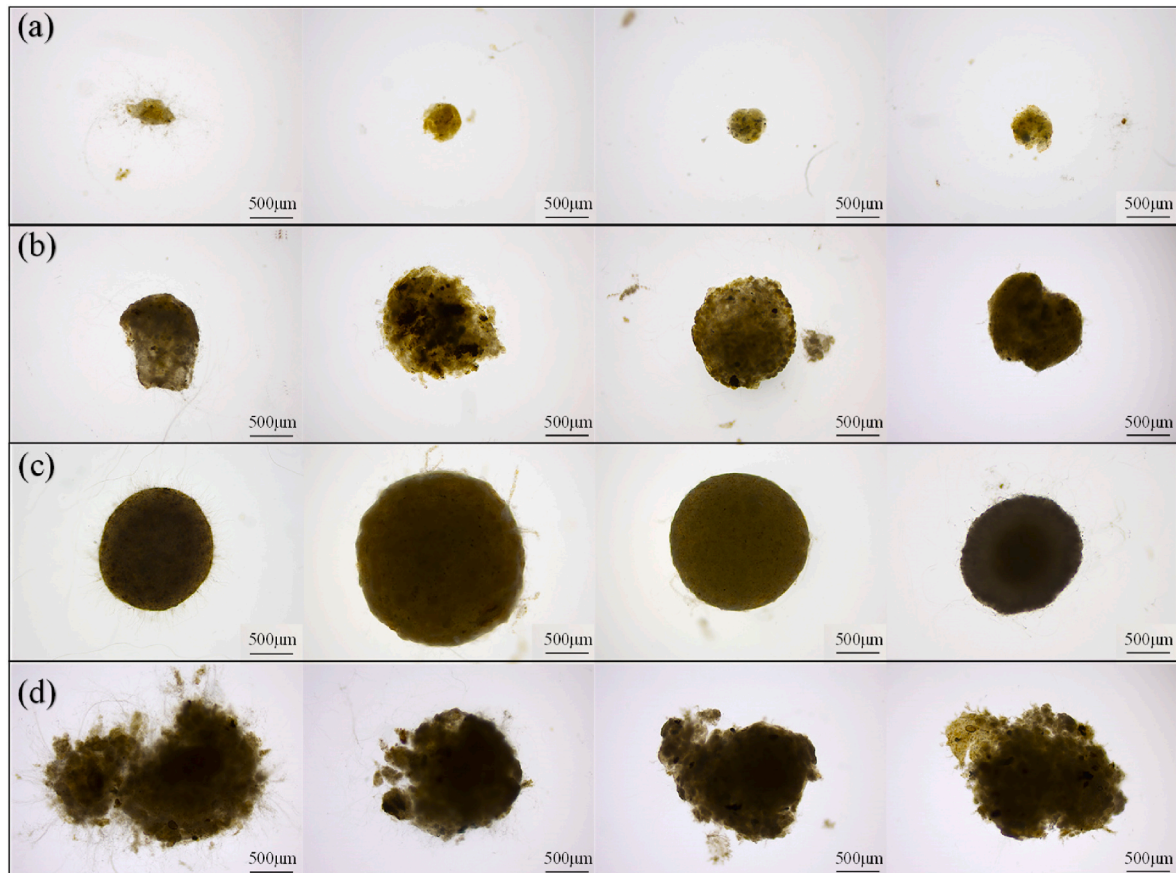


Fig. 1. Image of AGS over its whole life-cycle: (a) Initial stage; (b) Growth stage; (c) Maturity stage; (d) Cleavage stage.

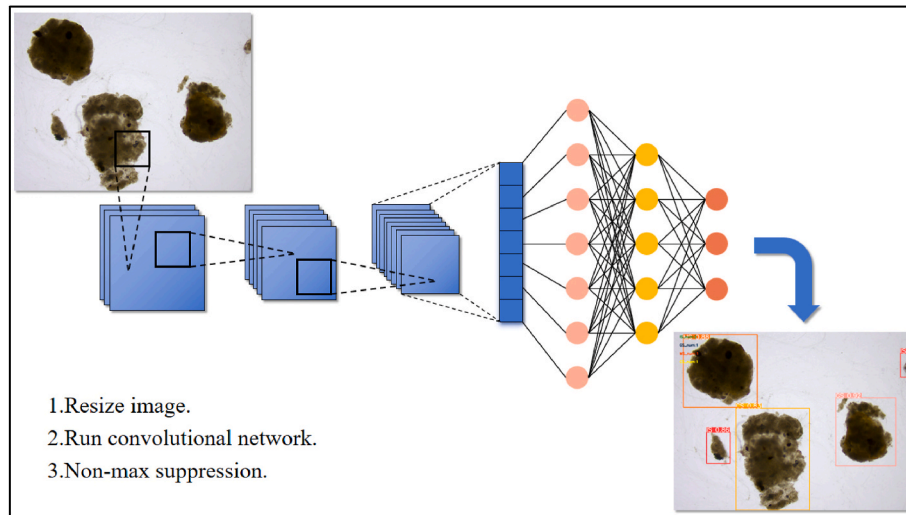


Fig. 2. The YOLO detection framework operates by: (1) adjusting the input image to a 640×640 resolution; (2) applying a convolutional network to the resized image; (3) filtering the detected results based on the confidence levels determined by the model.

terminated the training process if no improvement was observed during iteration. Additionally, in order to enhance prediction accuracy effectively, the Mosaic enhancement was turned off during the last 10 rounds of iteration in the data augmentation part of training (Alexey et al., 2020). During the forward propagation stage, the input samples were sequentially processed through each layer of the neural network until generating the predicted results. Once the neural network predictions were obtained, they were compared with the actual labels using a loss function to quantify their differences. By applying the chain rule, the gradients of the loss function were back-propagated from the output layer to input layer and further propagated back to the weights and biases at each layer. Once the back-propagation completed and the gradients for each parameter obtained, the network parameters could be updated using the gradient descent algorithm for the forward-backward propagation process in subsequent training iterations. Through continuous parameter adjustment and loss function minimization, the neural networks can gradually improve performance and accuracy. During the prediction stage, the Soft non-maximum suppression (soft-NMS) algorithm was employed to smoothly suppress the detection box results that exhibited higher overlap with the local maximum values (Navaneeth et al., 2017).

In order to enhance the efficiency of the experiment and practical application capabilities, this study utilized a cloud-based graphics processing unit (GPU) platform called Featurize. The hardware configuration included an RTX3060 graphics card with 12 GB of graphics memory and a Central Processing Unit (CPU) consisting of $6 \times$ E5 2680 v4, along with 28 GB of RAM and a 50 GB disk storage capacity. The runtime environment employed Python v3.10.12 and the PyTorch v2.0.1 machine learning library.

2.4. Model evaluation

For a more comprehensive understanding of the model's performance, it is crucial to load the test set in order to evaluate the effectiveness of the trained model. Hence, this study employed various evaluation indexes, including the confusion matrix, F1-score, precision and recall curve for analysis, in which, the confusion matrix, consisting of a 5×5 grid that included a background class, visually represented the relationship between actual and predicted values across different categories, together with the instances of mis-classification and false positives. Precision refers to the predictive accuracy in identifying the positive samples correctly, whereas recall measures how well positive samples are identified from the original dataset. The F1-score, ranging

from 0 to 1, serves as a balanced measure by combining both precision and recall into one metric, with the optimal performance indicated by 1, and the least effective by 0.

The procedures were implemented using selected dependencies from the scikit-learn library embedded in Python 3.10.12. Cross Entropy Loss (CE Loss), a key loss function in deep learning for classification tasks, quantified the discrepancy between the model's predicted probability distribution and the actual probability distribution of the label. In a multi-classification problem with C classes, the model predicted an input $P = \{p_1, p_2, \dots, p_C\}$, while the actual probability distribution was typically represented by a vector $Y = \{y_1, y_2, \dots, y_C\}$. When the model's predicted probability p_j for class j approached 1, the loss would be close to 0. Conversely, as p_j approached zero, the loss increased significantly according to Eq. (1) (Chicco et al., 2021).

$$CE\ Loss = - \sum_{i=1}^C y_i \log(p_i) \quad (1)$$

where, C represents the total number of categories, while p_i symbolizes the predicted probability for class i , while $y_i = 1$, if the true label corresponds to class i and 0 otherwise.

The Mean Squared Error (MSE) method is commonly employed to calculate the discrepancy between the predicted bounding boxes, characterized by coordinates (x, y) and dimensions (w, h) , and the actual bounding box (Chicco et al., 2021). This loss component evaluates the confidence error for each predicted bounding box with the aim of minimizing the mean square variance between the predicted and actual values (Eq. (2)).

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (2)$$

where, y_i represents the value of the i th sample for the target variable, $\hat{y}$ denotes the predicted value, and $\bar{y}$ signifies the mean of the aggregated samples.

2.5. Model interpretability

In order to provide a more comprehensive explanation of the model features, it is necessary to utilize interpretability analysis tools. The t-distributed Stochastic Neighbor Embedding (t-SNE) is a well-established technique in machine learning for reducing dimensionality, which can effectively transform data from a high-dimensional space to a lower-dimensional one while preserving the inherent information and

precise distribution characteristics during transmission, then furthermore, preserving the local characteristics of the dataset (Husnain et al., 2019). To achieve this, an appropriate amount of low-dimensional data was randomly generated, and the loss function measuring the difference between probability distributions was calculated. The gradient descent method was then employed to update the data until obtaining low-dimensional representations that meet specific requirements.

The similarity between each pair of data points in a high-dimensional space was initially computed using a Gaussian distribution, while the similarity between data points in a lower-dimensional space was calculated using a t-distribution. Kullback-Leibler divergence (KLD) serves as a loss function for minimizing the dissimilarity between the similarity distributions in high- and low-dimensional spaces, typically achieved through gradient descent optimization. For two points x_i and x_j within a high-dimensional dataset, Eq. (3) presents its definition.

$$p_{ji} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)} \quad (3)$$

where, p_{ji} represents the probability that x_j is adjacent to x_i , with higher values indicating closer they are. σ_i denotes the variance of the N nearest points to x_i , where N is a hyperparameter defined as perplexity. For two points y_i and y_j in a low-dimensional dataset, two definitions are provided as shown in (Eqs. (4)-(5)).

$$q_{ji} = \frac{\exp(-\|y_i - y_j\|^2)}{\sum_{k \neq i} \exp(-\|y_i - y_k\|^2)} \quad (4)$$

$$C = \sum_i \sum_j p_{ji} \log \frac{p_{ji}}{q_{ji}} \quad (5)$$

where, the meaning of q_{ji} is equivalent to that of p_{ji} . The low-dimensional data are randomly generated at the beginning, followed by the definition of a loss function based on p_{ji} and q_{ji} . C represents the Kullback-Leibler Divergence (KLD), which serves as the loss function minimized by t-SNE.

After importing the training model, the output results of the middle layer were extracted as the semantic features, then, loaded the test set image classification results and calculated the semantic features for each image, set the dimension and maximum number of iterations, and applied t-SNE to reduce the dimensionality to two dimensions. This approach effectively visualized clustering patterns throughout the AGS life-cycle and quantified the impact of each test set data feature on the overall dataset diversity. Additionally, for better interpretability of statistical data, the clustering results were visualized and further analyzed the internal feature correlations using the sklearn manifold library embedded in Python 3.10.12.

The SHapley Additive exPlanations (SHAP) method belongs to a category of additive feature attribution methods that is grounded in the amalgamation of game theory with local interpretation (Scott and Su-In, 2017), and its fundamental principle is the Shapley value, which considers both positive and negative contributions. In this model context, SHAP enables the elucidation of the influence of specific samples and features on the predicted outcome, facilitating the quantification of the local interpretations of the model. To simplify the input to z , SHAP operates by mapping x to z through the function $x = h_x(z)$. By utilizing z , the original model $f(x)$ can be approximated as a linear function of a binary variable (Eqs. (6)-(7)).

$$f(x) = g(z) = \phi_0 + \sum_{i=1}^M \phi_i z_i \quad (6)$$

where $z = \{0, 1\}^M$, M represents the count of input features, $\phi_0 = f(h_x(0))$, and ϕ_i is the feature attribution value.

$$\phi_i = \sum_{S \in F \setminus \{i\}} \frac{|S|!(M - |S| - 1)!}{M!} [f_x(S \cup \{i\}) - f_x(S)] \quad (7)$$

where F denotes the set of non-zero inputs in z , while S denotes a subset of F that excludes the i th feature. The term ϕ_i refers to the SHAP value, which is an aggregate metric for additive feature attributions. Currently, SHAP stands out due to its unique combination of properties including local accuracy, accounting for missing values, and consistency with its application (Scott et al., 2018).

The deep SHAP algorithm is utilized for interpreting the predictions made by deep learning models, in which, a baseline value, typically the average of input features or a zero-value matrix, is selected as a reference point to evaluate the relative contribution of each feature to the output, and assess how it impacts the output in relation to this baseline value. For deep neural networks, the interpretation proceeds layer by layer from input to output layers, enabling more precise quantification of the contribution of each node or feature. Ultimately, every input feature is assigned a SHAP value that quantifies its impact on the model's output (Wang et al., 2022).

After loading the categories of the classification dataset and organizing the tensor dimensions, the number of pixels and channels were obtained, which were then pre-processed to construct a model prediction function. The SHAP interpretability analysis algorithm was utilized to generate input images with distinct characteristics throughout the life-cycle of AGS. Considering both accuracy and computational efficiency, the batch size was set to 50, and the number of iterations was set at 8000, while a 64×64 px blur masking method was defined to mask the local area on the input image. By visualizing the heatmap, one could observe the contributions predicted by the model, where red indicated positive effects and blue represented negative effects. Moreover, darker colors indicated more significant effects. These procedures were also implemented using the SHAP library embedded in Python 3.10.12.

2.6. Prediction and statistics of AGS over its life-cycle

The prediction pattern was utilized to predict a new image using the trained model, which was loaded from a checkpoint file and provided with a fresh image for inference. To assess the actual predictive performance of this model and validate its generalization capability, the AGS images throughout their entire life-cycle were collected from another MBR and categorized into seven batches of more than 50 images each, resulting in 362 valid data points for analysis. Subsequently, the acquired microscopic images were predicted by setting an Intersection over Union (IoU) threshold of 0.7 to select the best detection outcome in the candidate boxes with considerable overlap, and these results were then saved in a designated directory. Additionally, global variables were defined to record the number of categories predicted for each input image, while batch statistics on the accumulated number of categories across iterations were used as functions. By implementing intelligent identification and automatic statistical functions, the role of this model in practical application scenarios has been significantly enriched, and the monitoring efficiency over the entire life-cycle of AGS can be greatly improved.

3. Results and discussion

3.1. Model training and testing

During the process of model training and testing, the box loss was used to monitor the regression of the detection box and predict the discrepancy between the detection and calibration boxes. In contrast, classification loss (cls loss) was employed to evaluate the classification accuracy and determine whether the anchor frame matched with its corresponding calibration classification. Additionally, distributional feature learning loss (dfl loss) was implemented to track the correlation

between different feature areas and probability distribution within the dataset. As the iteration count increased, these various loss metrics gradually decreased in value within the training set until they eventually converged to a single final curve. The results indicated that when Mosaic data enhancement was disabled during the last 10 epochs of training, there was a certain decrease in all types of losses, suggesting that this operation effectively improved model accuracy (Fig. 3(a–c)). The depicted results from model training demonstrated a comprehensive training stage with well-suited chosen model parameters (Fig. 3(d–f)).

The effectiveness of the detector is commonly assessed through the mean average precision (mAP) across various classes. In this study, the metric was derived from ranking detection scores specific to each class (Everingham et al., 2010), and the IoU measured the accuracy of object detection in a particular dataset, which was the ratio of the predicted frames to the actual frames in object detection. The mAP50 set IoU to 0.5, calculated AP for each class of images, and then averaged them across all classes, therefore, mAP50-95 could be understood as an average value calculated at the intervals of 0.05 from 0.5 to 0.95 at different IoUs. With an increase in iterations, there was a gradual improvement in precision, recall, mAP50, and mAP50-95 indicators, which initially fluctuated but eventually approached a plateau close to 1 (Fig. 3(g) and (h)). The final model achieved an mAP50 score of 0.985 (Fig. 3(i)), while the mAP50-95 reached 0.837 (Fig. 3(j)). In reality, Kong et al. utilized the YOLO model to identify microorganisms in microscopic images of activated sludge, bolstering its detection capabilities by incorporating deformable convolutional networks and attention mechanisms (Kong and Shen, 2023). Their model achieved an mAP50 score of 0.937. Notably, in this experiment, the mAP50 score reached an impressive 0.985, demonstrating that our model exhibits exceptionally high accuracy at an IoU threshold of 0.5. Overall, the evaluation results demonstrated that the model trained with the provided data exhibited effective performance leading to accurate prediction outcomes.

3.2. Confusion matrix

A confusion matrix primarily serves the purpose of juxtaposing the classification results with actual measurements, which can effectively illustrate the accuracy of the classification outcomes and is a widely used tool for visually evaluating the effectiveness of supervised learning algorithms. Moreover, the most commonly employed metric for assessing imbalanced classification performance is derived from the confusion matrix (Luque et al., 2019). In the resulting matrix shown in Fig. 4, the main diagonal line represents the proportion of correctly predicted values compared to their true counterparts, while other positions indicate different types of mis-classifications presented in a 5×5 format.

Background, as a category, refers to the background part of an image that is typically distinguished from the target or foreground. As illustrated in Fig. 4, both IS and CS images exhibited the highest correct prediction rate in the test set, reaching 0.97 each. However, approximately 0.03 images of IS and CS were misclassified as GS due to their similarity at the critical values since GS represents the subsequent stage of IS in its life-cycle. Additionally, when granule sizes were closely aligned, GS and CS displayed rough edges and appeared similar to each other. The correct prediction rate for GS images was 0.96 with around 0.04 GS images being mistakenly classified as CS for similar reasons mentioned earlier. On the other hand, MS images had the lowest correct prediction rate at 0.94, where approximately 0.02 MS images were wrongly identified as GS while another 0.04 MS images were misclassified as CS due to their resemblance with AGS before and after this particular stage of development, and smaller ones being predicted as GS while larger ones being labeled as CS. In all instances where the background was incorrectly predicted as the target, AGS accounted for the proportions of 0.38, 0.38, 0.15, and 0.08 across different stages of its life-cycle depending on granule size, indicating that larger granules occupied more image area, resulting in higher probability of being classified correctly. Research indicates that the improved YOLOv7 underwater object detection model achieves accuracy rates of 81% for conch species, 90% for sea urchin species, 89% for scallop species, and 89% for starfish species (Liu et al., 2023). Due to the particularities of

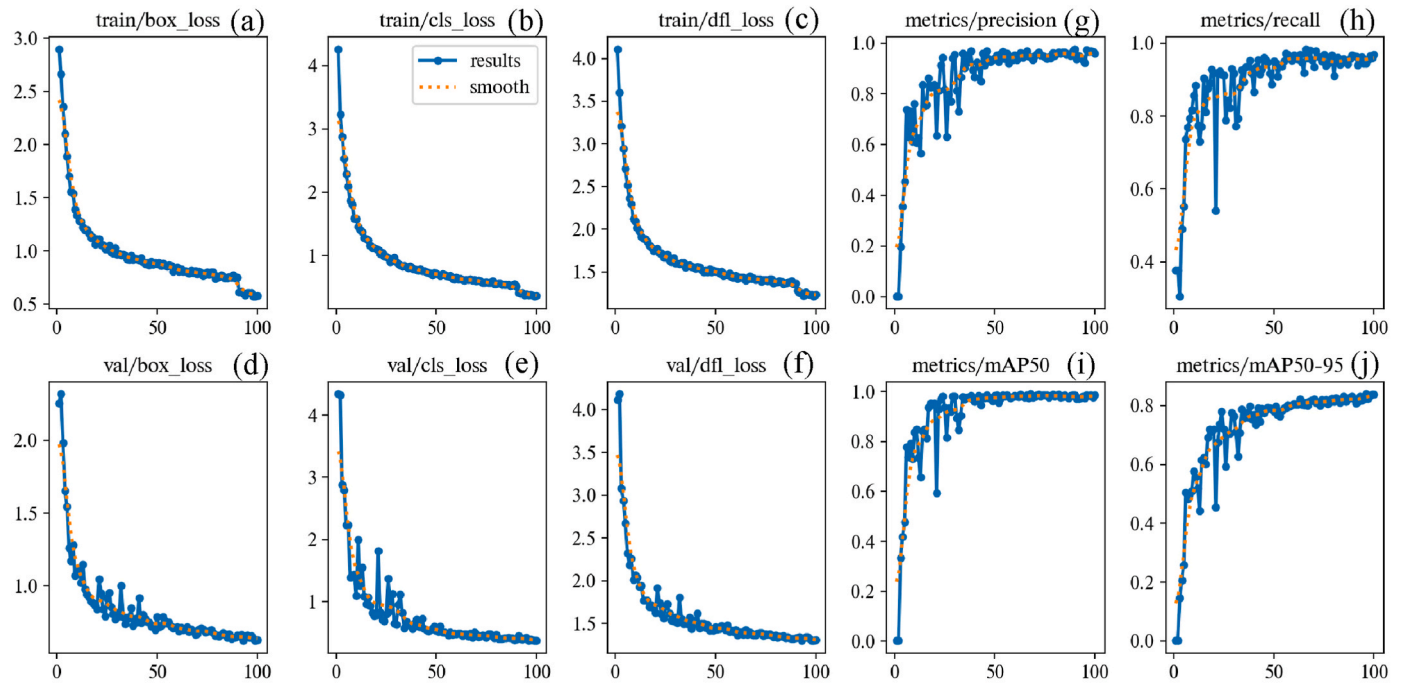


Fig. 3. The results of various loss functions and evaluation indexes in the process of 100 rounds of model iteration: (a), (b) and (c) are the results of iterations of box loss, cls loss and dfl loss on the training set, respectively; (d), (e) and (f) are the results of iterations of box loss, cls loss and dfl loss on the test set, respectively; (g) and (h) are the results of precision and recall on the test set; (i) and (j) are the results of mAP 50 and mAP50-95 on the test set.

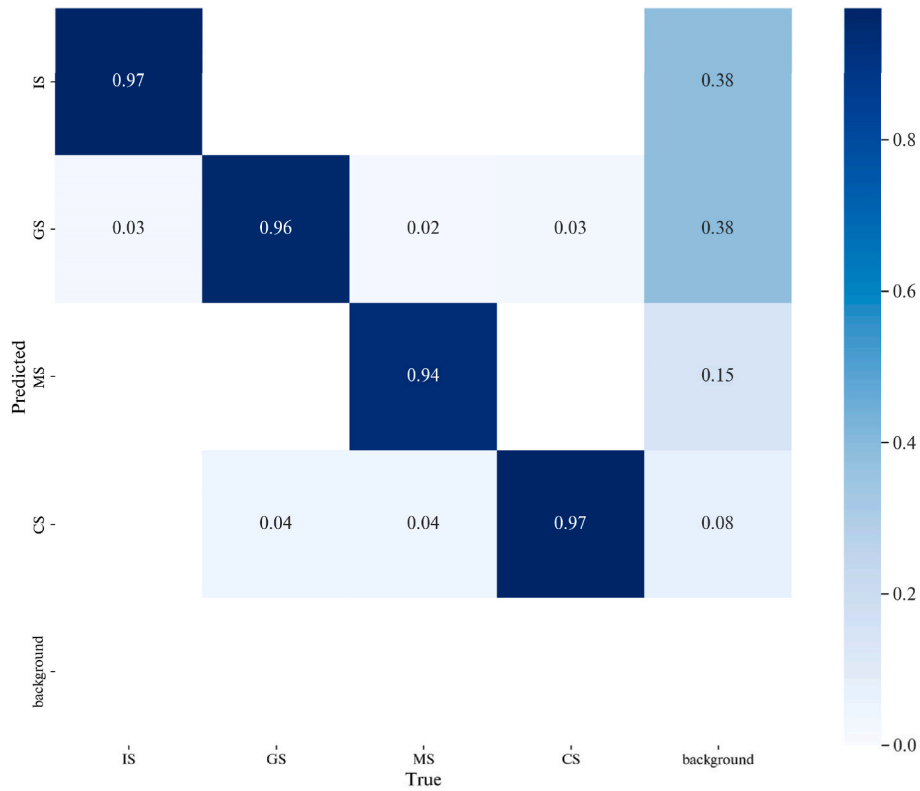


Fig. 4. Normalization results of all classes in the confusion matrix.

the underwater environment, such as poor lighting conditions, optical properties of water, and dynamically changing backgrounds, the distinction between targets and backgrounds becomes challenging. In this study, the interference from the background was minimal, hence the probability of AGS being correctly identified and classified was higher, which was also related to the standardized collection method of AGS micrographs.

3.3. Analysis of the model evaluation parameter

The precision rate measures the accuracy of a model's predictions for positive samples, that is, the proportion of true positives among the predicted positive samples. The recall rate reflects the model's ability to identify actual positive samples, which is the proportion of true positives among the actual positive samples. There is a trade-off between precision and recall; increasing precision may decrease recall, and vice versa. The confidence threshold is used to adjust this trade-off by setting a

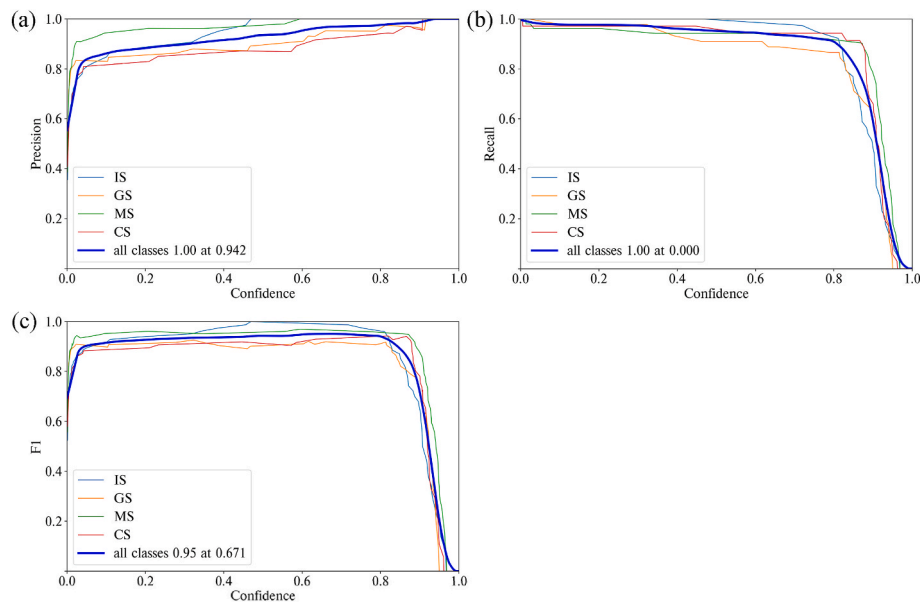


Fig. 5. Precision, recall and F1-score curves of AGS with different life-cycle under different confidence thresholds: (a) and (b) show the changing trends of precision and recall of various classes at different confidence levels, respectively; (c) shows the changing trend of F1-score.

standard for accepting prediction results. A high threshold increases precision but may decrease recall, while a low threshold increases recall but may decrease precision. The F1 score, as the harmonic mean of precision and recall, comprehensively evaluates model performance. Its value ranges from 0 to 1, with higher values indicating a better balance between precision and recall, thus avoiding performance bias caused by solely pursuing one of these metrics. In practical applications, an appropriate confidence threshold should be set according to the task requirements to achieve the desired precision and recall, and the overall performance of the model should be measured using the F1 score (Reda et al., 2020). The results of analyzing the model evaluation parameters are shown in Fig. 5.

When the confidence interval of IS and MS was 0.4–0.6, the precision gradually reached 1, while, the corresponding precision for GS and CS also gradually reached 1 for confidence interval of 0.8–1.0. Among them, IS showed the fastest growth rate, which indicated lower false detection rates for IS and MS features, making them easier to be accurately recognized by the model. In the confidence interval of 0.8–1.0, compared to GS and CS, IS was found to be more distinguishable from other AGS stages, indicating a lower false detection rate. This observation aligns with the fact that IS and MS represent relatively smaller-grained AGS targets, which are more susceptible to background interference and partial occlusion in images (Qian et al., 2022). For IS and MS, a precision of 1 could be achieved even with a confidence level of 0.4–0.6, suggesting that under certain specific conditions, models can still yield accurate detection results even with lower confidence levels. This highlights the importance of not solely relying on confidence levels to assess detection performance in practical applications, but rather integrating other metrics and actual scenarios for a comprehensive evaluation. However, when the confidence threshold was set at 0.942, all classes achieved the maximum value of 1 (Fig. 5(a)). As recall is generally negatively correlated with confidence levels, setting a threshold of 0 resulted in the maximum average recall for all classes (Fig. 5(b)). The smaller granule size and dense distribution of IS in images led to a higher recall rate for detecting smaller AGS granules when using low confidence intervals. When the confidence interval was set between 0.8 and 1.0, MS and CS had higher recall rates than IS and GS, with IS showing a significant decline. MS and CS, with their larger size and distinct features, resisted background noise and occlusion better, leading to higher recall rates. Conversely, IS had higher recall rates at lower confidence levels due to less stringent detection criteria, allowing more potential targets to be identified. However, at high confidence levels, IS's recall rate dropped significantly, indicating the model's limitation in recognizing smaller-grained IS targets, which are more affected by background interference and occlusion. This highlighted the need to balance the confidence threshold. Setting it too high increases false negatives, especially for small targets, while setting it too low increases false positives (Thatikonda et al., 2024). Thus, finding a balance between recall and false positive rates is crucial to optimize the model's performance. In summary, the higher recall rates for MS and CS at high confidence levels revealed the different characteristics of targets with varying granularities, while the significant drop in IS's recall rate emphasized the importance of balancing the confidence threshold in practical applications.

In this study, we determined that the confidence threshold should be set at 0.671. Setting the confidence threshold at 0.671 means that only predictions with a confidence level of 67.1% or higher were considered as positive samples. This threshold was used to balance precision and recall, ensuring that the model's predictions were both reliable and capable of capturing a sufficient number of actual positive instances. The average F1 score of 0.95 indicated that the model performed well in minimizing false positives and false negatives (Fig. 5(c)), thereby achieving a high level of overall performance. Therefore, it is recommended to use a confidence threshold of 0.671 when making predictions on new images to ensure optimal prediction accuracy.

3.4. Cluster analysis

The t-SNE technique, a machine learning method for visualizing and reducing the dimensionality of data, excels in preserving the integrity of relationships and structures in high dimensional data when it is projected onto a lower dimensional space (Pouyet et al., 2018). This technique is commonly employed to visually explore complex datasets and uncover patterns or clusters that may not be immediately discernible in the original high-dimensional space (Silva and Melo-Pinto, 2023). In this section, it was used to analyze the clusters for all classes with the results shown in Fig. 6.

As shown in Fig. 6, the features of AGS in the test set across its four different life-cycle stages were simplified to two dimensions and presented as four clusters. Overall, all types exhibited independent spatial distributions with low levels of coupling, indicating that the experimental classification criteria for the AGS life-cycle were reasonable. Specifically, CS showed the tightest distribution and a high degree of cohesion, suggesting a consistent pattern of morphological features. In contrast, GS exhibited the most dispersed distribution and showed specific coupling relationships with the other three categories. This implied that AGS, at various stages of development, diversified in appearance due to a variety of external conditions and internal developmental laws, leading to changes from light to dark colors and from rough to smooth edges. The corresponding complexity results confirmed that each class in the confusion matrix was predicted as GS. On the other hand, IS and MS only showed certain coupling relationships with adjacent life-cycle categories while maintaining similar morphological features within their respective clusters. This distribution pattern indicated that, despite certain interactions between adjacent stages, each stage retained unique characteristics identifiable through cluster analysis. The spatial independence and specific coupling patterns provided valuable insights into the dynamics of AGS development and confirmed the robustness of the classification method used in this study.

3.5. SHapley additive exPlanations (SHAP)

In the context of machine learning models, SHAP elucidates the impact of each sample and feature on the respective predicted value (Wang et al., 2024). By utilizing the SHapley value, it can quantify the local interpretation of the model, providing insights into the rationale behind the decision-making for a specific sample (Qi et al., 2025). In this section, it was first loaded the test images from each life-cycle, their dimensions were sorted out, and then a model prediction function was constructed after preprocessing. Considering the computation time and pixel granularity, an appropriate number of iterations were set to initialize the explainer, and the corresponding Shapley values were calculated for each category, test image, and pixel, and finally the heat map was generated as shown in Fig. 7.

The model primarily focused on capturing global features of small-grained images and edge features of large-grained images. As shown in Fig. 7, the global features in the IS and GS test images significantly influenced the model's predictions for MS. Specifically, the global features in IS and GS images, such as color variations and rough edges, provided the model with key information to distinguish between these two stages. The presence of these features enabled the model to more accurately identify the MS stage, as images in the MS stage typically had different colors and smoother edges. Furthermore, the edge features contained in the MS test images also had an adverse effect on the model's predictions for IS and GS. This indicated that there were significant morphological differences between IS, GS, and MS. IS and GS could be distinguished by their color variations and rough edges, while MS was characterized mainly by dark brown or black colors and smooth edges. This finding further validated the rationality of the aforementioned classification criteria. The internal structure and detailed features in the CS test images had a significant impact on the predictions for IS, as AGS fragments from CS re-emerged as IS AGS, thereby initiating a new

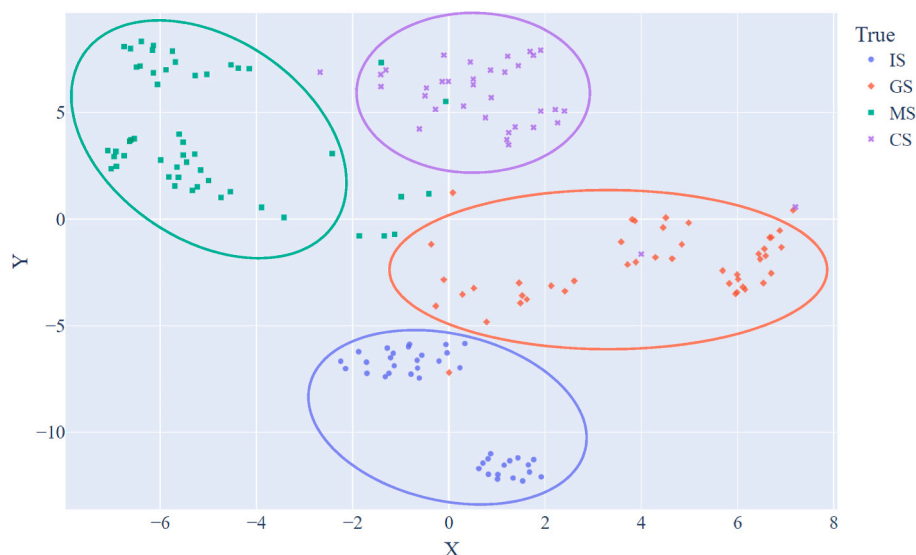


Fig. 6. The t-SNE cluster analysis of all classes.

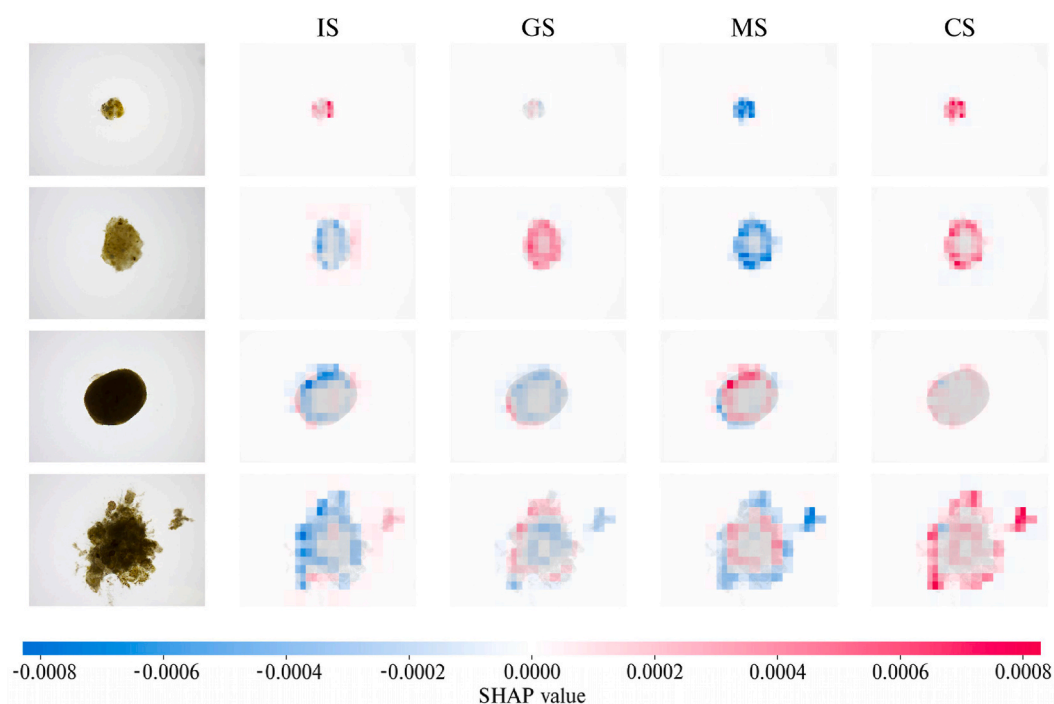


Fig. 7. Analysis of the SHAP feature interpretability of all classes.

life-cycle iteration. This process not only revealed the dynamic changes in the AGS life-cycle but also emphasized the model's capability in capturing key features of these changes.

In summary, the model captures key features of different stages, effectively distinguishing each life-cycle phase of AGS. These findings not only validate the rationality of the classification criteria but also provide a new perspective for understanding the life-cycle of AGS.

3.6. Model prediction and statistics

To validate the practical application of the model, images of AGS were collected from the AGS-MBR reactor over one week in the laboratory. Microscopic imaging was performed under consistent equipment and environmental conditions. The resulting images were then imported

into the model for prediction, as illustrated in Fig. 8.

The AGS with larger sizes, such as MS and CS, are more likely to be detected and less likely to be missed (Fig. 8(a)–(c)). Conversely, the AGS with smaller sizes, such as IS and GS, have a higher probability of being missed (Fig. 8(d)–(f)). When there is a certain degree of gap between the sludge granules, the prediction accuracy improves. However, if the gap is too small, instances of missed detection may occur. The prediction performance was enhanced under clean background conditions, but might be influenced by complex interference in the surrounding environment. The counting results were derived based on these predicted outcomes, and the various types could be counted in multiple batches, with the calculation time varying according to the computing power. In this model, preprocessing of each image required 2.6–3.4 ms, inference took 10.0–10.4 ms, and post-processing for statistical overlay consumed

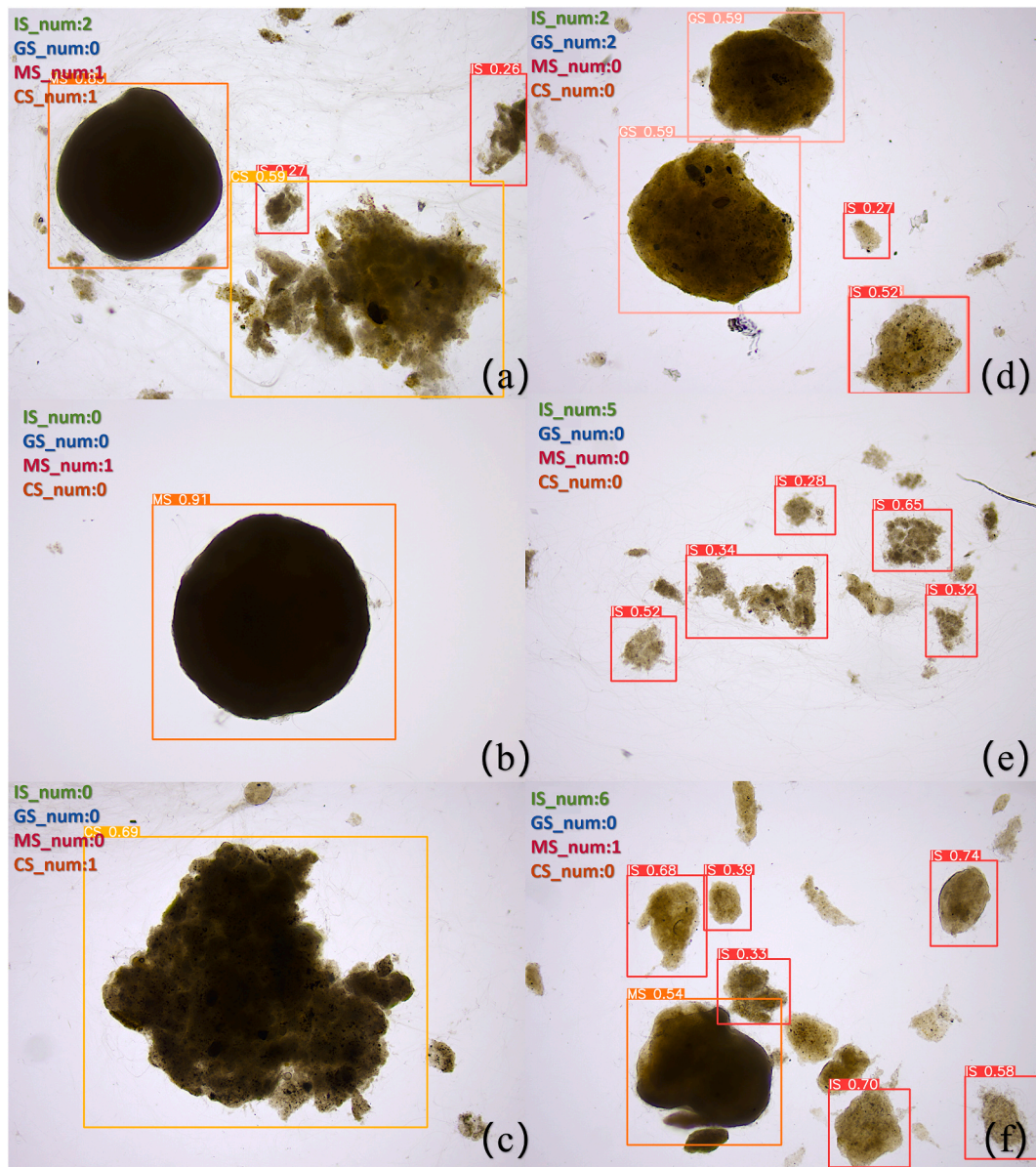


Fig. 8. The new AGS image import model was used to predict: (a), (b) and (c) are main predictions for some larger targets; (d), (e) and (f) are main predictions for some smaller targets and mixed-size targets.

1.6–1.8 ms, summing up to a total processing time of 14.2–15.6 ms. The model was used to develop statistical functions for performing analysis on AGS samples collected from the four previously mentioned stages, where each day's sample constituted a separate batch. The results are presented in Fig. 9.

As depicted in Fig. 9, the number of IS was the highest in each batch, with a total exceeding 700, followed by GS, while MS and CS exhibit the lowest number. This distribution suggested that the overall AGS was currently in a fragmentation phase, undergoing regeneration and gradual growth. Although IS had the highest count in the reactor, exceeding 700, the number alone could not serve as the sole criterion for determining the most dominant category, given that IS typically had a smaller granularity. In contrast, CS, being the category with the largest granularity, had a total count surpassing that of MS, indicating that the volume proportion of CS was certainly greater than that of MS. Therefore, it could be inferred that a significant portion of AGS in the reactor was currently in the CS state. The breakdown of CS into the IS state (Yang et al., 2023) likely accounted for the increased number of IS instances. At present, the number of MS in the MBR was not substantial, and it did

not represent a dominant state. Under these circumstances, managers should consider measures such as increasing the organic load and controlling hydraulic shear forces to promote AGS granulation, thereby increasing the proportion of MS AGS in the system (Lin et al., 2020). The statistical function of the model intuitively presents the quantity of each type of AGS in each batch and the total number of each type throughout the entire cycle, reflecting its real-time status in the MBR system. This provides valuable guidance for managers in selecting appropriate operating parameters and enhancing the maintenance efficiency of the MBR.

4. Conclusions

This study developed a YOLOv8-based machine learning model for identifying and classifying AGS life-cycle stages. Utilizing a standardized microscopy image acquisition method, the model achieved mAP50 of 0.985 and mAP50-95 of 0.837, indicating high precision and robustness. t-SNE analysis revealed distinct clustering of AGS features across stages, while SHAP analysis highlighted the model's proficiency in capturing

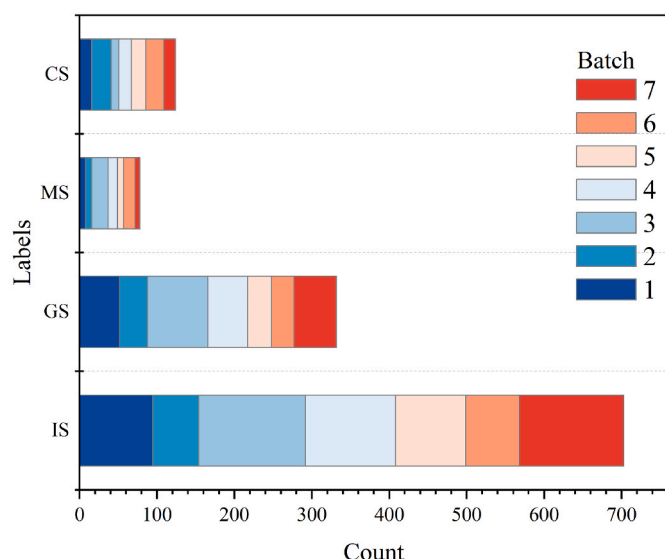


Fig. 9. The image statistics for each stage of the life-cycle of each AGS batch together with the aggregated total for all seven batches.

global features in fine-grained images and edge features in coarse-grained images, thereby validating the classification rationality. Enhanced with global variables and attention mechanisms, the model significantly improved detection efficiency and accuracy, enabling real-time AGS monitoring in MBR system and providing batch counting functionality for classification. The AGS in this study was cultured in a continuous-flow MBR system, whereas in SBR system, retraining the model and selecting appropriate YOLO modules for specific datasets are recommended for reliable detection. Additionally, while AI can effectively assess AGS status, translating these assessments into operational decisions requires extensive practical experience and consideration of multiple operational parameters. In summary, this study developed a YOLOv8-based machine learning model that provides an efficient and reliable solution for the automatic detection and classification of AGS life-cycles. These findings not only offer technical support for optimizing AGS-MBR system management but also pave new avenues for the application of artificial intelligence in wastewater treatment.

CRediT authorship contribution statement

Yu Zeng: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Chenguang Zhao:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation. **Danling Ma:** Investigation, Data curation. **Liyang Bin:** Methodology, Formal analysis. **Weirui Chen:** Resources. **Ping Li:** Software, Project administration. **Bing Tang:** Writing – original draft, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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